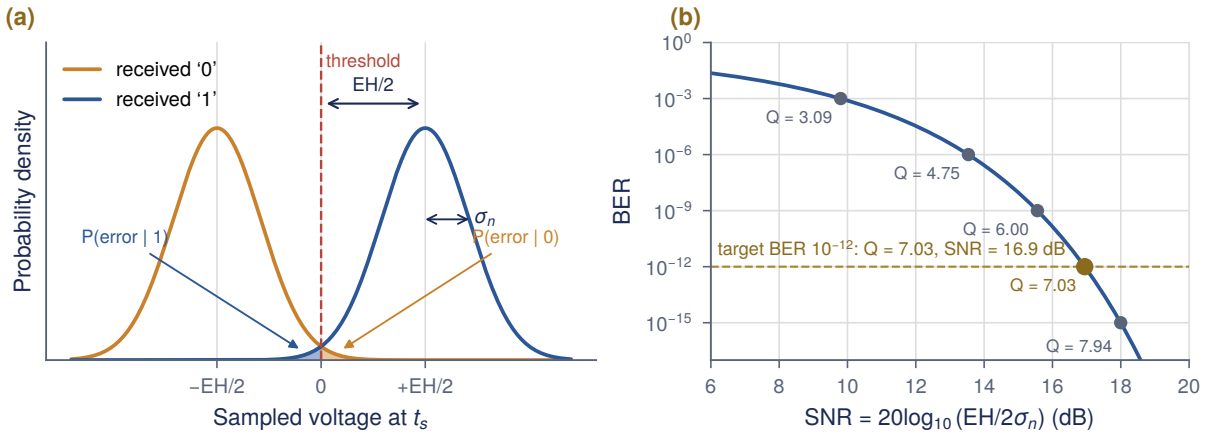
(a)
received '0'
received '1'
threshold
EH/2
Probability density
P(error | 1)
σ_n
P(error | 0)
−EH/2
0
+EH/2
Sampled voltage at t_s
(b)
BER
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
Q = 3.09
Q = 4.75
Q = 6.00
target BER 10^{-12}: Q = 7.03, SNR = 16.9 dB
Q = 7.03
Q = 7.94
6
8
10
12
14
16
18
20
SNR = $20\log_{10}(EH/2\sigma_n)$ (dB)